MECHTRON / SFWRENG 3K04: Software Development

Project Description – Fall 2026

McMaster University

Part B: Hospital Monitoring System

1 Project Context

This project focuses on the development of a software system for a centralized Monitoring System for Automated Dispensing Cabinets (ADCs). The Monitoring System is a safety-critical, human-facing coordination application that works with one or more ADCs.

The Monitoring System is one part of a larger hospital medication management solution. It provides centralized visibility of multiple ADC units and supports hospital personnel in monitoring inventory, alarms, cabinet status, and maintenance activities.

As a result, the Monitoring System is not the software that operates the physical dispensing mechanism. Instead, it works together with the separate ADC device-control software. While the ADC controls locks, slots, sensors, the motor/carousel, the dispensing actuator, and final local safety checks, the Monitoring System provides the broader user-facing, clinical-workflow, and administrative view needed by interested stakeholders.

Therefore, the ADC and the Monitoring System, while two separate software systems developed independently, must work together and be integrated later. The purpose of this integration activity is to allow two independently designed software systems to communicate and exchange information as would be expected in real-life application scenarios.

Regardless of the part each sub-team is assigned, sub-teams are expected to analyze the provided information, identify missing details, make reasonable assumptions, and develop system requirements and design solutions. Both Parts A and B focus on system analysis and design covering the software development lifecycle, not only software implementation.

2 Introduction

Hospitals operate multiple ADCs in different locations. As the number of cabinets increases, pharmacy personnel require a centralized system to monitor their operation and identify problems. A hospital plans to deploy a Monitoring System capable of communicating with multiple ADC units.

It is important to realize that the Monitoring System is safety-critical because incorrect behaviour or malfunction may lead to harm even though it does not physically release medication. Examples include authorizing the wrong user, approving the wrong dose session, sending a stale schedule, hiding or misrouting a critical alert, treating an offline device as healthy, sending a command to the wrong cabinet or slot, or assuming that an uncertain command succeeded, or incorrectly reporting that medication was administered.

In this project, the Monitoring System represents the application-side software layer responsible

for managing users and roles, maintaining medication schedules and device configuration, receiving device events, presenting dashboards and alerts, making or recording authorization decisions, coordinating refill and maintenance work, and preserving safety-relevant evidence.

The goal of Part B is to design the supervisory software system responsible for monitoring and managing ADC information. A secondary intended objective is to pair this Monitoring System with the ADC system through appropriate integration, which requires ADC teams and Monitoring System teams working together on the interface definitions for their proper communication.

3 Project Scope and Assumptions

As Part B focuses on a software solution responsible for supervising and coordinating with ADC units, it is important to understand its overall scope and assumptions.

3.1 Project Scope

The Monitoring System is responsible for managing:

- collecting information from ADC units;
- monitoring cabinet status;
- managing alarms;
- storing historical information;
- generating reports;
- providing operator dashboards;
- monitoring cabinet health;
- responding to requests received from ADC units;
- distributing configuration and operational information to ADC units.

The following items are outside the project scope:

- physical ADC hardware;
- direct control of cabinet components;
- internal design of ADC software;
- hospital patient information systems;
- medication decision processes;
- network infrastructure design.

Sub-teams working on this part should focus on the Monitoring System software considering its interaction with ADC units and users.

3.2 Project Assumptions

Sub-teams should assume that:

- multiple ADC units exist;
- ADC units provide communication interfaces;
- hospital network infrastructure is available;
- users have valid accounts;
- ADC operational information is available through communication interfaces.

Any additional assumptions and decisions should be documented.

4 System Overview

As mentioned before, the Monitoring System is a supervisory software application that communicates with multiple ADC units. Unlike the ADC, the Monitoring System does not directly interact with physical hardware.

The Monitoring System manages information related to:

- cabinet operation;
- inventory;
- alarms;
- maintenance;
- system health.

The system receives information from ADC units and provides centralized monitoring functions for hospital personnel. The detailed design of ADC software is intentionally outside the scope of this project.

A simple block diagram showing the system overview is depicted in Figure 1.

5 Operational Environment

The Monitoring System operates within the hospital information environment.

Typical users include:

- nurses;
- pharmacists;
- technicians;

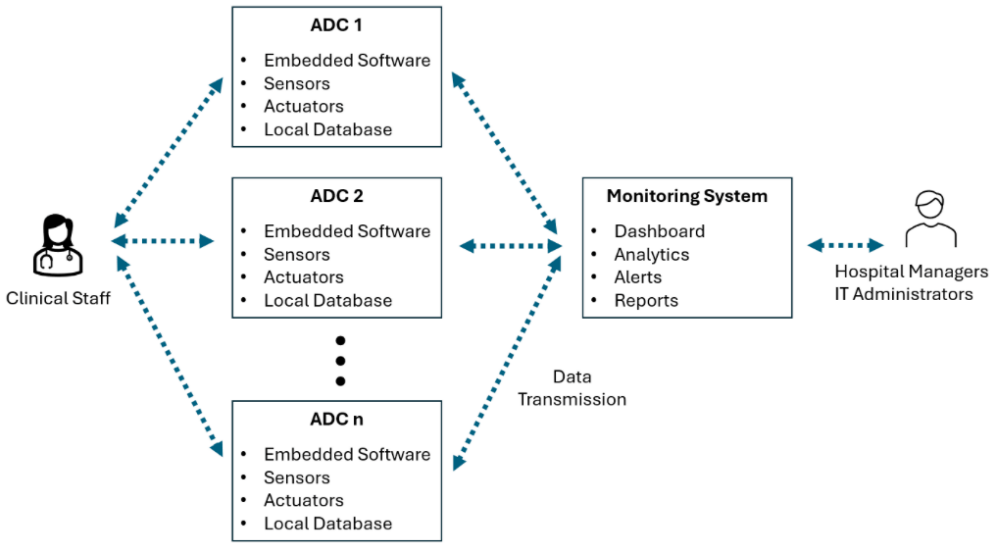


Figure 1: Simple block diagram showing the system overview.

- administrators.

The system must operate in an environment where:

- information from multiple ADC units must be managed and reported;
- operational events must be available for review;
- alarms must be identified;
- system availability must be monitored.

6 High-Level System Capabilities

The Monitoring System is expected to support the following capabilities, intentionally described in high level terms. Your team must use them as reference points for the development of a software solution.

- **ADC Communication Management:** The system should communicate with multiple ADC units.
- **Inventory Supervision:** The system should monitor medication inventory information received from ADCs.
- **Alarm Management:** The system should receive and manage alarms generated by ADC units.
- **Historical Data Management:** The system should store operational information for later review.
- **Reporting:** The system should provide reports related to ADC operation.

- **Operator Dashboard:** The system should provide users with centralized visibility of ADC information.
- **Cabinet Health Monitoring:** The system should monitor ADC availability and health information.
- **Request Handling:** The system should respond to requests received from ADC units.

7 Example Operational Scenarios

7.1 Scenario 1: Inventory Monitoring

A pharmacist reviews inventory information from multiple ADC units. The Monitoring System identifies cabinets that require attention.

7.2 Scenario 2: Alarm Reception

An ADC sends an alarm message to the Monitoring System. The system receives, records, and displays the alarm information.

7.3 Scenario 3: ADC Communication Failure

One ADC stops communicating. The Monitoring System detects the communication problem and updates the cabinet status.

7.4 Scenario 4: Historical Review

A pharmacist reviews previous dispensing activities from an ADC. The Monitoring System retrieves stored information and displays the requested data.

8 System Constraints

The Monitoring System design should consider the following constraints:

- The system must communicate with multiple ADC units.
- Information received from ADCs must remain consistent.
- Important events must be stored.
- User access must be controlled.
- Communication problems must be detected.
- Historical information must remain available.
- The system must support future expansion.

9 External Interfaces

The Monitoring System communicates with ADC units. Information exchanged may include:

- operational status;
- inventory information;
- alarms;
- maintenance information;
- system health information.

The interface requirements between Monitoring System and ADC are intentionally not provided. A few integration considerations are provided below to help teams think about this problem.

9.1 Integration Considerations

The Monitoring System team will work with an ADC team to define the communication interface between both systems.

The Monitoring System team is responsible for defining:

- information received from ADCs;
- information sent to ADCs;
- communication requirements;
- data storage and synchronization requirements;
- handling of unavailable ADC systems.

The final goal is to demonstrate that the Monitoring System can communicate correctly with independently developed ADC software.